

## Problem 1 - Fourier Series

Compute the coefficients  $a_n, b_n$  in the Fourier series representations of each of the following functions. Then, determine the  $x$  at which the series representations fail, if any.

$$a_n = \frac{1}{L} \int_{-L}^L f(x) \sin\left(\frac{n\pi}{L}x\right) dx$$
$$b_n = \frac{1}{L} \int_{-L}^L f(x) \cos\left(\frac{n\pi}{L}x\right) dx$$

(a)  $f(x) = 1 + \cos(x) - \sin(2x) \quad L = \pi$

$$\tilde{f}(x) = \frac{b_0}{2} + \sum_{n=1}^{\infty} a_n \sin(nx) + b_n \cos(nx)$$
$$1 + \cos(x) - \sin(2x) = \frac{b_0}{2} + \sum_{n=1}^{\infty} a_n \sin(nx) + b_n \cos(nx)$$

By inspection:

$$\frac{b_0}{2} = 1 \implies b_0 = 2$$
$$b_1 = 1, \quad b_n = 0 \text{ for } n \neq 1$$
$$a_2 = -1, \quad a_n = 0 \text{ for } n \neq 2$$

Since  $f(x) = 1 + \cos(x) - \sin(2x)$  is smooth and continuous, its Fourier series representation does not fail for any  $x$ .

(b)  $f(x) = x, \quad L = \pi$

Solving for  $a_n$  and  $b_n$  using formulas from class.

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} x \sin(nx) dx$$

Integration by parts (table method):

|      |                           |
|------|---------------------------|
| $x$  | $\sin(nx)$                |
| $-1$ | $-\frac{1}{n} \cos(nx)$   |
| $0$  | $-\frac{1}{n^2} \sin(nx)$ |

$$\begin{aligned}
a_n \pi &= -\frac{x}{n} \cos(nx) + \frac{1}{n^2} \sin(nx) \Big|_{-\pi}^{\pi} \\
a_n \pi &= -\frac{\pi}{n} \cos(n\pi) + \frac{1}{n^2} \sin(n\pi) - \left( -\frac{-\pi}{n} \cos(-n\pi) + \frac{1}{n^2} \sin(-n\pi) \right) \\
a_n \pi &= -\frac{\pi}{n} \cos(n\pi) - \left( -\frac{-\pi}{n} \cos(-n\pi) \right) \\
a_n &= -\frac{1}{n} \cos(n\pi) - \left( -\frac{1}{n} \cos(-n\pi) \right) \\
a_n &= -\frac{1}{n} \cos(n\pi) - \left( \frac{1}{n} \cos(-n\pi) \right) \quad \cos(n\pi) = \cos(-n\pi) = (-1)^n \\
a_n &= -\frac{1}{n} (-1)^n - \left( \frac{1}{n} (-1)^n \right) \\
a_n &= -\frac{2}{n} (-1)^n
\end{aligned}$$

Since  $f(x) = x$  is an odd function, and  $\cos(nx)$  is an even function,  $f(x) \cos(nx)$  is an odd function. The integral  $b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} x \cos(\frac{n\pi}{\pi} x) dx$  would be 0 since it is an odd function integrated over symmetric bounds. Therefore  $b_n = 0 \forall n \in \mathbb{Z}^+$

The series fails at  $x = \pi$  and  $x = -\pi$  because of the jump discontinuities that exist there.

(c)  $f(x) = e^x$ ,  $L = \pi$

Using the formula for  $a_n$ :

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} e^x \sin(n\pi) dx$$

Using integration by parts with  $u = \sin(nx)$ ,  $du = n \cos(nx)$ ,  $dv = e^x$ ,  $v = e^x$

$$\int_{-\pi}^{\pi} e^x \sin(n\pi) dx = \sin(nx)e^x - n \int e^x \cos(nx)$$

Using integration by parts again with  $u = \cos(nx)$ ,  $du = -n \sin(nx)$ ,  $dv = e^x$ ,  $v = e^x$

$$\begin{aligned} \int_{-\pi}^{\pi} e^x \sin(n\pi) dx &= \sin(nx)e^x - n \left( \cos(nx)e^x + n \int e^x \sin(nx) dx \right) \\ \int_{-\pi}^{\pi} e^x \sin(n\pi) dx &= \sin(nx)e^x - n \cos(nx)e^x - n^2 \int e^x \sin(nx) dx \\ \int_{-\pi}^{\pi} e^x \sin(n\pi) dx + n^2 \int_{-\pi}^{\pi} e^x \sin(nx) dx &= \sin(nx)e^x - n \cos(nx)e^x \Big|_{-\pi}^{\pi} \\ \int_{-\pi}^{\pi} e^x \sin(n\pi) dx (1 + n^2) &= \sin(nx)e^x - n \cos(nx)e^x \Big|_{-\pi}^{\pi} \\ \int_{-\pi}^{\pi} e^x \sin(n\pi) dx &= \frac{\sin(nx)e^x - n \cos(nx)e^x}{1 + n^2} \Big|_{-\pi}^{\pi} \end{aligned}$$

We know that  $\cos(n\pi) = \cos(-n\pi) = (-1)^n$

$$\begin{aligned} \int_{-\pi}^{\pi} e^x \sin(n\pi) dx &= \frac{-n(-1)^n e^{\pi}}{1 + n^2} + \frac{n(-1)^n e^{-\pi}}{1 + n^2} \\ \int_{-\pi}^{\pi} e^x \sin(n\pi) dx &= \frac{-n(-1)^n}{1 + n^2} (e^{\pi} - e^{-\pi}) \\ a_n &= \frac{-n(-1)^n}{1 + n^2(\pi)} (e^{\pi} - e^{-\pi}) \end{aligned}$$

For  $b_n$ , we can also use the formula:

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} e^x \cos(nx) dx$$

Using integration by parts, with  $u = \cos(nx)$ ,  $du = -n \sin(nx)$ ,  $v = e^x$ ,  $dv = e^x$ .

$$\int_{-\pi}^{\pi} e^x \cos(nx) dx = e^x \cos(nx) + n \int e^x \sin(nx)$$

Using integration by parts again with  $u = \sin(nx)$ ,  $du = -n \cos(nx)$ ,  $v = e^x$ ,  $dv = e^x$ .

$$\begin{aligned}
\int_{-\pi}^{\pi} e^x \cos(nx) dx &= e^x \cos(nx) + n \sin(nx) - n^2 \int e^x \cos(nx) \\
\int_{-\pi}^{\pi} e^x \cos(nx) dx + n^2 \int_{-\pi}^{\pi} e^x \cos(nx) dx &= e^x \cos(nx) + n \sin(nx) \Big|_{-\pi}^{\pi} \\
\int_{-\pi}^{\pi} e^x \cos(nx) dx (1 + n^2) &= e^x \cos(nx) + n \sin(nx) \Big|_{-\pi}^{\pi} \\
\int_{-\pi}^{\pi} e^x \cos(nx) dx &= \frac{e^x \cos(nx) + n \sin(nx)}{1 + n^2} \Big|_{-\pi}^{\pi} \\
\int_{-\pi}^{\pi} e^x \cos(nx) dx &= \frac{e^{\pi} \cos(n\pi)}{1 + n^2} - \frac{e^{-\pi} \cos(-n\pi)}{1 + n^2} \\
\int_{-\pi}^{\pi} e^x \cos(nx) dx &= \frac{e^{\pi} (-1)^n}{1 + n^2} - \frac{e^{-\pi} (-1)^n}{1 + n^2} \\
\int_{-\pi}^{\pi} e^x \cos(nx) dx &= \frac{(-1)^n}{1 + n^2} (e^{\pi} - e^{-\pi}) \\
b_n &= \frac{1}{\pi} \frac{(-1)^n}{1 + n^2} (e^{\pi} - e^{-\pi})
\end{aligned}$$

The series fails at  $x = \pi$  and  $x = -\pi$  because of the jump discontinuities that exist there.

$$(d) \quad f(x) = \begin{cases} 0, & -\pi \leq x \leq 0, \\ 1 - \frac{x}{\pi}, & 0 < x \leq \pi \end{cases}, \quad L = \pi$$

Since the function is 0 on the interval  $-\pi \leq x \leq 0$ , our bounds for our integrals only have to go from 0 to  $\pi$ .

For  $b_0$ :

$$\begin{aligned}
b_n &= \frac{1}{\pi} \int_0^{\pi} \left(1 - \frac{x}{\pi}\right) dx \\
&= \frac{1}{\pi} \left[ x - \frac{x^2}{2\pi} \right]_0^{\pi} \\
&= \frac{1}{\pi} \left[ \pi - \frac{\pi^2}{2\pi} \right] \\
b_0 &= 1 - \frac{1}{2} = \frac{1}{2}
\end{aligned}$$

For  $a_n$ :

$$a_n = \int_0^\pi \left(1 - \frac{x}{\pi}\right) \sin(nx) \, dx$$

Integration by parts using the table method:

|                     |                           |
|---------------------|---------------------------|
| $1 - \frac{x}{\pi}$ | $\sin(nx)$                |
| $\frac{1}{\pi}$     | $-\frac{1}{n} \cos(nx)$   |
| 0                   | $-\frac{1}{n^2} \sin(nx)$ |

$$\begin{aligned} a_n &= \frac{1}{\pi} \left[ -\left(1 - \frac{x}{\pi}\right) \frac{1}{n} \cos(nx) - \frac{1}{\pi n^2} \sin(nx) \right]_0^\pi \\ &= 0 - \left(\frac{-1}{n\pi}\right) a_n = \frac{1}{n\pi} \end{aligned}$$

For  $b_n$ :

$$b_n = \frac{1}{\pi} \int_0^\pi \left(1 - \frac{x}{\pi}\right) \cos(nx) \, dx$$

We can use integration by parts with the table method again:

|                     |                           |
|---------------------|---------------------------|
| $1 - \frac{x}{\pi}$ | $\cos(nx)$                |
| $\frac{1}{\pi}$     | $\frac{1}{n} \sin(nx)$    |
| 0                   | $-\frac{1}{n^2} \cos(nx)$ |

$$\begin{aligned} b_n &= \frac{1}{\pi} \left[ \left(1 - \frac{x}{\pi}\right) \frac{1}{n} \sin(nx) - \frac{1}{\pi n^2} \cos(nx) \right]_0^\pi \\ &= \frac{1}{\pi} \left[ -\frac{1^n}{n^2 \pi} + \frac{1}{n^2 \pi} \right] \\ b_n &= \frac{1 - (-1)^n}{\pi^2 n^2} \end{aligned}$$

The series fails at  $x = 0$  due to the jump discontinuity that exists there.

## Problem 2 - Another Separation of Variables IBVP

Solve the finite-interval, **free-endpoint** wave equation IBVP given by

(a) Solve

$$\begin{cases} u_{tt} = c^2 u_{xx}, & x \in [0, 3], t \geq 0, \\ u_x(0, t) = 0, & t \geq 0, \\ u_x(3, t) = 0, & t \geq 0, \\ u(x, 0) = \cos^4\left(\frac{2}{3}\pi x\right), & x \in [0, 3], \\ u_t(x, 0) = 0, & x \in [0, 3]. \end{cases}$$

$$u_x(0, t) = u_x(3t) = 0$$

$$u(x, t) = \frac{b_0}{2} + \sum_{n=1}^{\infty} b_n \cos\left(\frac{n\pi}{3}x\right) \cos\left(\frac{cn\pi}{3}t\right)$$

$$u(x, 0) = \cos^4\left(\frac{2}{3}\pi x\right)$$

$$\text{Let } \theta = \frac{2}{3}\pi x$$

$$\cos^4(\theta) = (\cos^2(\theta))^2 = \left(\frac{1}{2}(1 + \cos(2\theta))\right)^2 \frac{1}{4}(1 + 2\cos(2\theta) + \cos^2(2\theta))$$

$$\frac{1}{4}(1 + 2\cos(2\theta) + \frac{1}{2} + \frac{1}{2}\cos(4\theta))$$

$$\frac{3}{8} + \frac{1}{2}\cos(2\theta) + \frac{1}{8}\cos(4\theta)$$

$$u(x, 0) = \frac{3}{8} + \frac{1}{2}\cos\left(\frac{4}{3}\pi x\right) + \frac{1}{8}\cos\left(\frac{8}{3}\pi x\right)$$

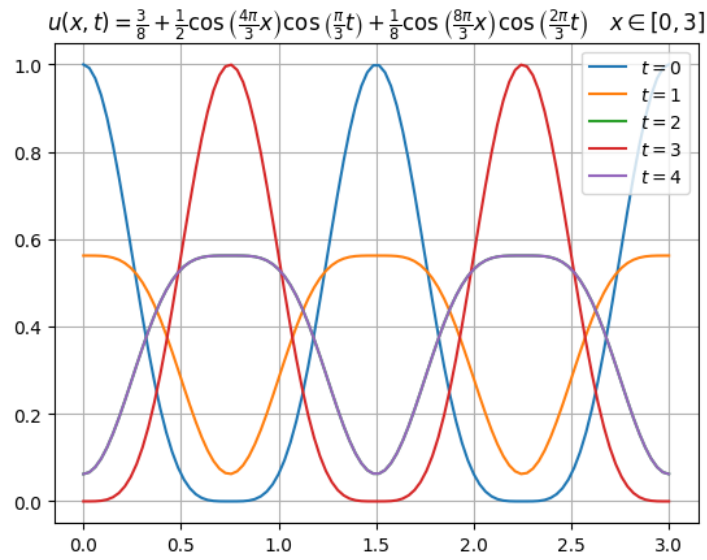
From inspection, we can see that:

$$\begin{aligned} n = 4, \quad b_4 &= \frac{1}{2} \\ n = 8, \quad b_8 &= \frac{1}{8} \\ n = 0, \quad b_0 &= \frac{3}{4} \end{aligned}$$

So the final equation is:

$$u(x, t) = \frac{3}{8} + \frac{1}{2} \cos\left(\frac{4\pi}{3}x\right) \cos\left(\frac{4c\pi}{3}t\right) + \frac{1}{8} \cos\left(\frac{8\pi}{3}x\right) \cos\left(\frac{8c\pi}{3}t\right)$$

(b) Plot



### Problem 3 - Laplace's Equation in Polar Coordinates

Laplace's equation,  $\Delta u = \nabla \cdot \nabla u = 0$ , appears in a huge variety of applications, including but not limited to fluid mechanics, finance, electrostatics, heat conduction, image processing, and acoustics. In polar coordinates, it is given by

$$u_{rr} + \frac{1}{r}u_r + \frac{1}{r^2}u_{\theta\theta} = 0,$$

(a) Use the separation of variables ansatz and derive the corresponding pair of ODEs

The ansatz is  $u(r, \theta) = R(r)\Theta(\theta)$ .

Calculating the partial derivatives:

$$\begin{aligned}u_r &= R'(r)\Theta(\theta) \\u_{rr} &= R''(r)\Theta(\theta) \\u_{\theta\theta} &= R(r)\Theta''(\theta)\end{aligned}$$

Plugging back into  $u_{rr} + \frac{1}{r}u_r + \frac{1}{r^2}u_{\theta\theta} = 0$ :

$$R''(r)\Theta(\theta) + \frac{1}{r}R'(r)\Theta(\theta) + \frac{1}{r^2}R(r)\Theta''(\theta) = 0$$

Multiply by original ansatz  $R(r)\Theta(\theta)$

$$\frac{R''(r)}{R(r)} + \frac{R'(r)}{rR(r)} + \frac{\Theta''(\theta)}{r^2\Theta(\theta)} = 0$$

Multiply by  $r^2$

$$\frac{r^2 R''}{R} + \frac{r R'}{R} + \frac{\Theta''}{\Theta} = 0$$

Separate. Since one side depends only on  $\theta$  and the other only on  $r$ , they must equal some constant  $\lambda$

$$\frac{r^2 R'' + r R'}{R} = -\frac{\Theta''}{\Theta} = \lambda$$

Simplify into two ODEs:

$$\begin{aligned}\Theta'' + \lambda\Theta &= 0 \\ r^2 R'' + rR' - \lambda R &= 0\end{aligned}$$

**(b) Solve the ODEs**

$$\begin{aligned}\Theta'' + \lambda\Theta &= 0 \\ r^2 R'' + rR' - \lambda R &= 0\end{aligned}$$

Starting with  $\Theta'' + \lambda\Theta = 0$ . Since  $\theta$  is  $2\pi$  periodic, the following conditions apply:

$$\begin{aligned}\Theta(\theta) &= \Theta(\theta + 2\pi) \\ \lambda &= n^2, \quad n = 1, 2, 3, \dots\end{aligned}$$

**Case  $\lambda = 0$**

$$\begin{aligned}\Theta'' &= 0 \\ \text{Integrating twice} \\ \Theta &= A_0 + B_0\theta \\ B_0 &= 0 \text{ since it is periodic, it is not allowed to grow. So:} \\ \Theta(\theta) &= A_0\end{aligned}$$

**Case  $\lambda > 0, \lambda = n^2$**

$$\begin{aligned}\Theta'' + n^2\Theta &= 0 \\ \text{Harmonic Oscillator} \\ \Theta &= B_n \cos(n\theta) + A_n \sin(n\theta)\end{aligned}$$

Solving  $r^2 R'' + rR' - n^2 R = 0$  next using cauchy euler:

$$\begin{aligned}R(r) &= r^\alpha \\ R'(r) &= \alpha r^{\alpha-1} \\ R'' &= \alpha(\alpha-1)r^{\alpha-2}\end{aligned}$$

Substituting back into our ODE:

$$\begin{aligned} r^2 R'' + r R' - n^2 R &= 0 \\ r^2(\alpha(\alpha - 1)r^{\alpha-2}) + r\alpha r^{\alpha-1} - n^2 r^\alpha &= 0 \\ (\alpha^2 - n^2) &= 0 \implies \alpha = \pm n \end{aligned}$$

**Case  $n = 0$ . (repeated root  $\alpha = 0$ )**

$$R_0(r) = C_0 + D_0 \ln(r)$$

**Case  $n > 0$ . (two roots,  $\alpha = n$  and  $\alpha = -n$ )**

$$R_n(r) = C_n r^n + D_n r^{-n}$$

Final ODE solutions:

$$\begin{aligned} \Theta &= B_n \cos(n\theta) + A_n \sin(n\theta) \\ R_n(r) &= C_n r^n + D_n r^{-n} \end{aligned}$$

**(c) Solve the boundary-value problem**

$$\begin{cases} u_{rr} + \frac{1}{r}u_r + \frac{1}{r^2}u_{\theta\theta} = 0, & 0 \leq r \leq R, 0 \leq \theta \leq 2\pi, \\ u(R, \theta) = \cos(\theta) \sin(\theta), & 0 \leq \theta < 2\pi \\ u(r, 2\pi) = u(r, 0), & 0 \leq r < R. \\ \lim_{r \rightarrow 0} |u(r, \theta)| < +\infty, & 0 \leq \theta < 2\pi. \end{cases}$$

*The 3rd and 4th constraints effectively state that the angular component is  $2\pi$  periodic and that  $u$  is bounded. Both of these conditions are necessary for the solution to be physical!*

$$\begin{aligned} \Theta &= B_n \cos(n\theta) + A_n \sin(n\theta) \\ R_n(r) &= C_n r^n + D_n r^{-n} \end{aligned}$$

Because  $\lim_{r \rightarrow 0} |u(r, \theta)| < +\infty$ ,  $r^{-n} = \frac{1}{r^n} \rightarrow \infty$  as  $r \rightarrow 0$ , so we need to get rid of that term in our second ODE by setting  $D_0 = 0$ ,  $D_n = 0$ .

$$\begin{aligned} \Theta &= B_n \cos(n\theta) + A_n \sin(n\theta) \\ R_0(r) &= C_0 \\ R_n(r) &= C_n r^n \end{aligned}$$

Now we can multiply our ODES together to get the general solution.

$$u(r, \theta) = \frac{b_0}{2} + \sum_{n=1}^{\infty} r^n (b_n \cos(n\theta) + a_n \sin(n\theta))$$

We can apply the initial condition  $u(R, \theta) = \cos(\theta) \sin(\theta)$

$$\begin{aligned} \frac{b_0}{2} + \sum_{n=1}^{\infty} R^n (b_n \cos(n\theta) + a_n \sin(n\theta)) &= \cos(\theta) \sin(\theta) \\ \frac{b_0}{2} + \sum_{n=1}^{\infty} R^n (b_n \cos(n\theta) + a_n \sin(n\theta)) &= \frac{1}{2} \sin(2\theta) \end{aligned}$$

By inspection:

$$\frac{b_0}{2} = 0 \implies b_0 = 0$$

$$\text{No cos terms} \implies b_n = 0$$

$$a_2 R^2 = \frac{1}{2} \implies a_2 = \frac{1}{2R^2}$$

**Final solution after subbing back into the general solution at  $n = 2$**

$$u(r, \theta) = \frac{r^2}{2R^2} \sin(2\theta)$$

## Problem 4 - Operators and Inner products

(a) Show that  $\langle u, v \rangle_{L^2}$  defines an inner product.

### 1. Symmetry

Since multiplication is commutative:

$$\langle u, v \rangle_{L^2} = \int_{-1}^1 u(x)v(x) dx = \int_{-1}^1 v(x)u(x) dx = \langle u, v \rangle_{L^2}$$

So it is symmetrical.

### 2. Linearity

$$\begin{aligned}\langle au + bv, w \rangle_{L^2} &= \int_{-1}^1 (au(x) + bv(x))w(x) dx \\ &= \int_{-1}^1 au(x)w(x) + bv(x)w(x) dx \\ &= \int_{-1}^1 au(x)w(x) dx + \int_{-1}^1 bv(x)w(x) dx \\ \langle au + bv, w \rangle_{L^2} &= a \langle u, w \rangle_{L^2} + b \langle v, w \rangle_{L^2}\end{aligned}$$

Thus,  $\langle u, v \rangle_{L^2}$  is linear.

### 3. Positive definiteness

Let  $u \in C_0^2([-1, 1], \mathbb{R})$  where  $u \neq 0$ :

$$\begin{aligned}\langle u, u \rangle_{L^2} &= \int_{-1}^1 u(x)u(x) dx \\ &= \int_{-1}^1 u(x)^2 dx\end{aligned}$$

$u(x)^2$  is always  $> 0$  so its integral from  $-1$  to  $1$  will also be  $> 0$ , thus  $\langle u, u \rangle_{L^2}$  is strictly positive.

- (b) Show that  $\mathcal{L} = \partial_x^2$  is self-adjoint with respect to  $\langle \cdot, \cdot \rangle_{L^2}$  on  $C_0^2([-1, 1], \mathbb{R})$ .

$$\begin{aligned}\langle \mathcal{L}u, v \rangle_{L^2} &= \int_{-1}^1 \partial_x^2 u(x) v(x) dx \\ &= \int_{-1}^1 u''(x) v(x) dx\end{aligned}$$

Using integration by parts with  $w = v(x)$ ,  $dw = v'(x)$ ,  $dz = u''(x)$ ,  $z = u'(x)$

$$\langle \mathcal{L}u, v \rangle_{L^2} = u'(x)v(x) \Big|_{-1}^1 - \int_{-1}^1 u'(x)v'(x) dx$$

Since  $u, v \in C_0^2([-1, 1], \mathbb{R})$ , we know  $v(1) = v(-1) = 0$ , so the boundary evaluates to 0.

$$\langle \mathcal{L}u, v \rangle_{L^2} = - \int_{-1}^1 u'(x)v'(x) dx$$

Using integration by parts again with  $w = u'(x)$ ,  $dw = u''(x)$ ,  $dz = v'(x)$ ,  $z = v(x)$ :

$$\langle \mathcal{L}u, v \rangle_{L^2} = - \left( u(x)v'(x) \Big|_{-1}^1 - \int_{-1}^1 u(x)v''(x) dx \right)$$

$u(1) = u(-1) = 0$  so the boundary evaluates to 0

$$\begin{aligned}\langle \mathcal{L}u, v \rangle_{L^2} &= \int_{-1}^1 u(x)v''(x) dx \\ &= \int_{-1}^1 u(x)\partial_x^2 v(x) dx \\ \langle \mathcal{L}u, v \rangle_{L^2} &= \langle u, \mathcal{L}v \rangle_{L^2}\end{aligned}$$

Thus,  $\mathcal{L} = \partial_x^2$  is self-adjoint.

- (c) Show that the eigenvalues of a self-adjoint operator are purely real and that the eigenvectors corresponding to different eigenvalues are orthogonal.

### 1. Purely Real Eigenvalues

Let  $\mathcal{L}v = \lambda v$  for some  $v \neq 0$ . Taking the inner product:

$$\langle \mathcal{L}v, v \rangle = \langle \lambda v, v \rangle = \lambda \langle v, v \rangle$$

By the definition of a self adjoint operator:

$$\begin{aligned} \langle \mathcal{L}v, v \rangle &= \langle v, \mathcal{L}v \rangle = \langle v, \lambda v \rangle = \bar{\lambda} \langle v, v \rangle \\ \lambda \langle v, v \rangle &= \bar{\lambda} \langle v, v \rangle \end{aligned}$$

Since  $v \neq 0$ ,  $\langle v, v \rangle > 0$ . Therefore  $\lambda = \bar{\lambda}$  meaning  $\lambda$  is purely real.

## 2. Orthogonality

Let  $\mathcal{L}v_1 = \lambda_1 v_1$  and  $\mathcal{L}v_2 = \lambda_2 v_2$  with  $\lambda_1 \neq \lambda_2$ .

$$\langle \mathcal{L}v_1, v_2 \rangle = \langle \lambda_1 v_1, v_2 \rangle = \lambda_1 \langle v_1, v_2 \rangle$$

Using the self adjoint property and the fact that  $\lambda_2$  is real:

$$\begin{aligned} \langle \mathcal{L}v_1, v_2 \rangle &= \langle v_1, \mathcal{L}v_2 \rangle = \langle v_1, \lambda_2 v_2 \rangle = \lambda_2 \langle v_1, v_2 \rangle \\ \lambda_1 \langle v_1, v_2 \rangle &= \lambda_2 \langle v_1, v_2 \rangle \\ (\lambda_1 - \lambda_2) \langle v_1, v_2 \rangle &= 0 \end{aligned}$$

Since  $\lambda_1 \neq \lambda_2$ ,  $\langle v_1, v_2 \rangle = 0$ , meaning the eigenvectors are orthogonal.

**(d) Show that  $\mathcal{L} = \partial_x^2$  is negative definite with respect to  $\langle \cdot, \cdot \rangle_{L^2}$  on  $C_0^2([-1, 1], \mathbb{R})$ .**

Showing that  $\langle \mathcal{L}u, u \rangle < 0$  for  $u \neq 0$ .

$$\langle \mathcal{L}u, u \rangle = \int_{-1}^1 u''(x)u(x) dx$$

Using integration by parts with  $w = u(x)$ ,  $dw = u'(x)$ ,  $dz = u''(x)$ ,  $z = u'(x)$ :

$$\langle \mathcal{L}u, u \rangle = u'(x)u(x) \Big|_{-1}^1 - \int_{-1}^1 u'(x)u'(x) dx$$

Since  $u \in C_0^2([-1, 1], \mathbb{R})$ ,  $u(1) = u(-1) = 0$ , so the boundary is 0.

$$\langle \mathcal{L}u, u \rangle = - \int_{-1}^1 (u'(x))^2 dx$$

Since  $(u'(x))^2 \geq 0$ , the integral is positive. Thus:

$$- \int_{-1}^1 (u'(x))^2 dx < 0 \implies \langle \mathcal{L}u, u \rangle < 0$$

So  $\mathcal{L}$  is negative definite.

(e) **Show that a negative-definite operator cannot have any positive eigenvalues.**

Let  $\lambda$  be an eigenvalue of  $\mathcal{L}$  with corresponding eigenfunction  $v \neq 0$ , such that  $\mathcal{L}v = \lambda v$ . Taking the inner product:

$$\langle \mathcal{L}v, v \rangle = \langle \lambda v, v \rangle = \lambda \langle v, v \rangle$$

Since  $\mathcal{L}$  is negative definite, we know  $\langle \mathcal{L}v, v \rangle < 0$ . Because inner products are positive definite, we know  $\langle v, v \rangle > 0$ .

$$\lambda = \frac{\langle \mathcal{L}v, v \rangle}{\langle v, v \rangle}$$

A negative number divided by a positive number is negative. Therefore,  $\lambda < 0$ , which means it cannot be positive.

(f) **Find all eigenvalues and associated eigenfunctions of  $\mathcal{L} = \partial_x^2$  on  $C_0^2([-1, 1], \mathbb{R})$ .**

$$v''(x) = \lambda v(x) \implies v''(x) - \lambda v(x) = 0$$

From part (e),  $\lambda < 0$ . Let  $\lambda = -n^2$  where  $n > 0$ .

$$v''(x) + n^2 v(x) = 0$$

Harmonic oscillator ODE, with general solution:

$$v(x) = A \cos(nx) + B \sin(nx)$$

Applying boundary conditions  $v(-1) = 0$  and  $v(1) = 0$ :

$$v(-1) = A \cos(-n) + B \sin(-n) = A \cos(n) - B \sin(n) = 0$$

$$v(1) = A \cos(n) + B \sin(n) = 0$$

$$2A \cos(n) = 0 \implies A = 0 \text{ or } \cos(n) = 0$$

$$2B \sin(n) = 0 \implies B = 0 \text{ or } \sin(n) = 0$$

**Case 1:  $B = 0$  and  $\cos(n) = 0$**

$$n = \frac{k\pi}{2} \text{ for } k = 1, 3, 5, \dots$$

**Case 2:**  $A = 0$  and  $\sin(n) = 0$

$$n = \frac{k\pi}{2} \text{ for } k = 2, 4, 6, \dots$$

With both cases, we get  $n = \frac{k\pi}{2}$  for  $k = 1, 2, 3, \dots$ . The eigenvalues are:

$$\lambda_k = -n^2 = -\left(\frac{k\pi}{2}\right)^2, \quad k = 1, 2, 3, \dots$$

The eigenfunctions are:

$$v_k(x) = \sin\left(\frac{k\pi}{2}(x+1)\right), \quad k = 1, 2, 3, \dots$$

## 5. Feedback

It took me a really long time to do this homework, even with 1 late token and the extension from canvas being down. Maybe it has to do with the fact that this is the first assignment I decided to write up using LaTeX (I hope I can get faster).

Question 4 did not feel really relevant to what we are doing in class, and I felt it was not a good use of my time having to do it.